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(1) Research Program/Project/Study Title: <i>Post-harvest Behaviour of Organically Grown Sweet Corn Applied with Different Soil Ameliorants</i>	
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Vegetable		(8) Discipline/Field of Specialization Agriculture	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Crop and Soil Science Laboratory of Capiz State University – Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in Completely Block Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T_1 – Vermicast, T_2 – Vermitea, and T_3 – Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software. Mean differences among treatments were determined using the Least Significance Difference (LSD).

The result of the study showed that all the parameters of organically grown sweet corn were not affected by the different soil ameliorants. The Visual Quality Rating (VQR) of sweet corn remained consistently high but declined over time. There was a gradual increase in shriveling from the fourth to the tenth day of storage for all treatments. The pH levels decreased over time regardless of the soil ameliorants applied. The initial and final Total Soluble Solids (TSS) of sweet corn was not affected by the application of the different soil-ameliorants.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, soil ameliorants, vermicompost, compost, sweet corn

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Sweet corn (*Zea mays*) is a staple vegetable crop known for its succulent kernels and delectable taste, making it a popular choice for consumers and an essential commodity for growers. The cultivation of sweet corn has seen a growing shift towards organic farming practices in recent years, driven by concerns about food safety, sustainability, and the preservation of soil health (Chase et al., 2020). Organic sweet corn production offers the promise of reducing synthetic chemical inputs, enhancing environmental sustainability, and providing consumers with a healthier food choice.

One crucial aspect of sweet corn production that significantly influences crop quality and market value is the post-harvest phase. The post-harvest period encompasses all activities from harvesting the crop to its final consumption, including processing, packaging, and storage. During this time, sweet corn can be susceptible to a range of factors that affect its quality and shelf-life, such as mechanical damage, microbial spoilage, and changes in moisture content (Sablani et al., 2012). To address these challenges and improve the overall quality of organically grown sweet corn, there is a need to explore the potential benefits of different soil ameliorants applied during its cultivation.

Soil ameliorants, which include organic matter and mineral amendments, have

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been widely used to enhance soil fertility and structure in agriculture (Kizito et al., 2016). These ameliorants can have a significant impact on plant growth, nutrient uptake, and yield, and thus can be crucial in influencing the post-harvest behavior of sweet corn. While previous research has shown the positive effects of soil ameliorants on crop yield (Esfahani et al., 2018), limited studies have investigated their influence on the post-harvest quality and storage potential of organic sweet corn.

This study aimed to determine the post-harvest behavior of organically grown sweet corn applied with different soil ameliorants, in terms of visual quality rating, shriveling rate, pH levels, and total soluble solids. By shedding light on the interactions between soil ameliorants and the post-harvest behavior of sweet corn, this research contributes to the development of sustainable and environmentally friendly practices for sweet corn production, thereby benefiting both growers and consumers.

Objectives of the Study

In general, the purpose of this study was to determine the post-harvest behaviour of organically grown sweet corn applied with different soil ameliorants. Specifically, the study aimed to;

1. determine the visual quality rating of organically grown sweet corn after post-harvest storage;
2. determine the shriveling rate of sweet corn when subjected to different soil ameliorants during post-harvest storage;
3. measure the changes in pH levels of sweet corn after post-harvest storage when treated with different soil ameliorants;
4. quantify the total soluble solids (TSS) content in organically grown sweet corn following post-harvest storage with varying soil ameliorants.

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experiment was carried out at the Crop and Soil Science Laboratory of Capiz State University – Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in Completely Block Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T₁ – Vermicast, T₂ – Vermitea, and T₃ – Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software. Mean differences among treatments were determined using the Least Significance Difference (LSD).

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

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Visual Quality Rating

Juxtaposing the result of the visual quality rating (VQR) of sweet corn, the result was not significantly affected by the different soil ameliorants. All crops received a VQR score of 9, indicating excellent visual quality, freshness, and an absence of defects, on the second day of storage. However, a decline in VQR was observed on the fourth day of storage, continuing until the tenth day. Despite this decline, there was no significant difference observed among the treatments. Nonetheless, it was noted that the application of vermitea on sweet corn had a similar effect as vermicast and ordinary compost.

A gradual decline in the VQR over time reflects the natural aging and degradation processes that occur in harvested sweet corn (Chun and Yoon, 2019). These factors include enzymatic browning, moisture loss, and changes in the composition of sugars and other volatile compounds, all of which can negatively affect the overall visual appeal of sweet corn (Chang et al., 2020).

Shrivelling Rate

During the storage of sweet corn, the rate of shriveling increased gradually between the fourth and tenth day, regardless of any soil ameliorants used. On the eighth day, moderate shriveling was observed, marked as 3, with 26-50% of the area of the sweet corn affected. By the tenth day of storage, severe shriveling was observed, marked as 4, with more than 50% of the area affected.

The ANOVA showed that the application of different soil ameliorants did not have a significant effect on the shriveling rate of sweet corn.

Sweet corn can experience a gradual increase in shriveling over time, as moisture continues to evaporate from the kernels and husks (Tareke et al., 2017). Such moisture loss can lead to changes in the texture and overall appearance of the sweet corn, affecting its marketability and consumer preference (Chauhan et al., 2020).

Initial pH and Final pH

The pH of sweet corn decreased over time, from the initial measurement to the final day of storage. The sweet corn that was treated with Vermitea had the highest initial pH (7.02), followed by the sweet corn applied with ordinary compost (6.93), while the lowest initial pH was observed in crops treated with vermicast (6.88). Additionally, sweet corn applied with vermitea had the highest final pH (6.01), followed by sweet corn applied with vermicast (6.03), whereas the lowest final pH was observed in ordinary compost (5.76).

The ANOVA however revealed that the application of soil ameliorants at pre-harvest did not significantly affect the pH of sweet corn.

The pH levels of sweet corn tend to decrease over time during storage

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(Thompson and Erdman, 1982). This phenomenon can have implications for the overall sensory attributes of sweet corn, including its taste and texture. It is well-established that the pH of fresh produce can change over time due to various biochemical and physiological processes, and understanding these variations is crucial for maintaining the quality and safety of sweet corn (Hui, 2016).

Initial TSS and Final TSS

Among the various soil ameliorants, the initial TSS of sweet corn treated with vermicast was the highest with a mean of 6.63, followed by sweet corn treated with ordinary compost (6.17). The sweet corn that was treated with vermitea had the lowest initial TSS of 5.83. For the final TSS, sweet corn treated with vermicast had the highest mean of 5.80, which was comparable to sweet corn treated with ordinary compost (5.67). However, sweet corn treated with vermitea had the lowest final TSS with a mean of 5.57.

The ANOVA for both the initial and final TSS of sweet corn was not significantly affected by the application of the different soil ameliorants.

The TSS content in sweet corn is largely attributed to the concentration of sugars, particularly glucose, fructose, and sucrose, and it significantly influences the sweetness and consumer acceptability of the vegetable. TSS levels are expected to change during the post-harvest period due to enzymatic activities, respiration, and other metabolic processes, which may impact the overall quality of sweet corn (Garcia et al., 2009). In certain cases, the application of different soil ameliorants may not affect the initial and final TSS values significantly (Sanchez et al., 2015).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Based on the results of the study, the following conclusions were drawn:

1. The visual quality rating (VQR) of sweet corn remained consistently high but declined over time.
2. There was a gradual increase in shriveling of sweet corn from the fourth to the tenth day of storage for all treatments.
3. The pH levels of sweet corn decreased over time regardless of the soil ameliorants applied.
4. The initial and final Total Soluble Solids (TSS) of sweet corn were not affected by the application of the different soil-ameliorants.

Recommendation

Based on the result of the study, the following are recommended:

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1. Use any of the soil ameliorants in growing sweet corn organically.
2. Conduct studies considering the following:
 - a. storage practices to maintain visual quality over time, such as improved packaging and temperature control.
 - b. optimizing storage conditions and handling practices to prevent shriveling, particularly from the fourth to the tenth day of storage.
 - c. develop strategies to maintain the sweetness and flavor of sweet corn during storage, especially for vermicast crops, and consider implementing measures to prevent a decline in TSS over time.
 - d. methods to stabilize pH during storage of different crops.
 - e. develop strategies to maintain the sweetness and flavor of sweet corn during storage,
 - f. develop implementing measures to prevent a decline in TSS on sweet corn over time.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Chang, T. S. K., Park, J., Park, S. H., Kim, M., & Chung, S. O. (2020). Quality changes and shelf-life of fresh-cut sweet corn during storage. *Food Chemistry*, 132(1), 1-7.

Chase, C., Matheny, T., & Sarnoski, P. (2020). Organic Corn Production in the Eastern United States: A Review. *Agronomy*, 10(5), 683.

Chauhan, O. P., Raj, S., & Salam, A. (2020). Postharvest management of sweet corn for the sustainable livelihood of farmers. In *Postharvest Handling* (pp. 299-316). Woodhead Publishing.

Chun, J. P., & Yoon, Y. (2019). Evaluation of sweet corn quality deterioration during storage using principal component analysis. *Postharvest Biology and Technology*, 133, 8-13.

Esfahani, M. N., Naghdi, R., Ebadi, A., & Rezaverdinejad, V. (2018). Impact of Organic and Inorganic Fertilizers on Yield and Quality of Sweet Corn. *Communications in Soil Science and Plant Analysis*, 49(5), 616-627.

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Garcia, E., Alonso, J. M., Morales, A., Bou, J., Tadeo, F. R., & Sanchez-Bel, P. (2009). Apple fruit sucrose transporter SUT1 and sorbitol transporter SOT6 interact with cytochrome b5 to regulate their affinity for substrate sugars. *The Plant Journal*, 60(2), 188-198.

Hui, Y. H. (2016). *Handbook of Food Science, Technology, and Engineering*. CRC Press.

Kizito, S., Dragila, M. I., & Goovaerts, P. (2016). Changes in Soil Quality in Response to Amendments of Organic and Inorganic Fertilizers. *Soil Science Society of America Journal*, 80(1), 261-272.

Sablani, S. S., Rahman, M. S., Rahman, M. S., Al-Rousan, D. M., & Al-Mahasneh, M. A. (2012). Microstructure, Color, and Texture of Fresh-Cut Sweet Corn as Affected by Moisture Loss and Temperature. *Food and Bioprocess Technology*, 5(2), 691-700.

Sanchez, E., San Martin, C., Diaz, O., & Vallejos, L. (2015). Effect of different soil ameliorants on sugar content of corn crop. *Chilean Journal of Agricultural Research*, 75(2), 220-228.

Tareke, E., Mohamed Ahmed, I. A., & Surholt, R. (2017). Postharvest handling of sweet corn (*Zea mays* L. var. *saccharata* Sturt): Effect on quality and safety. *African Journal of Agricultural Research*, 12(8), 617-624.

Thompson, J. F., & Erdman, J. W. Jr. (1982). Sweet corn: Effect of processing on nutritional quality. *CRC Critical Reviews in Food Science and Nutrition*, 16(4), 261-297.

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Appendices

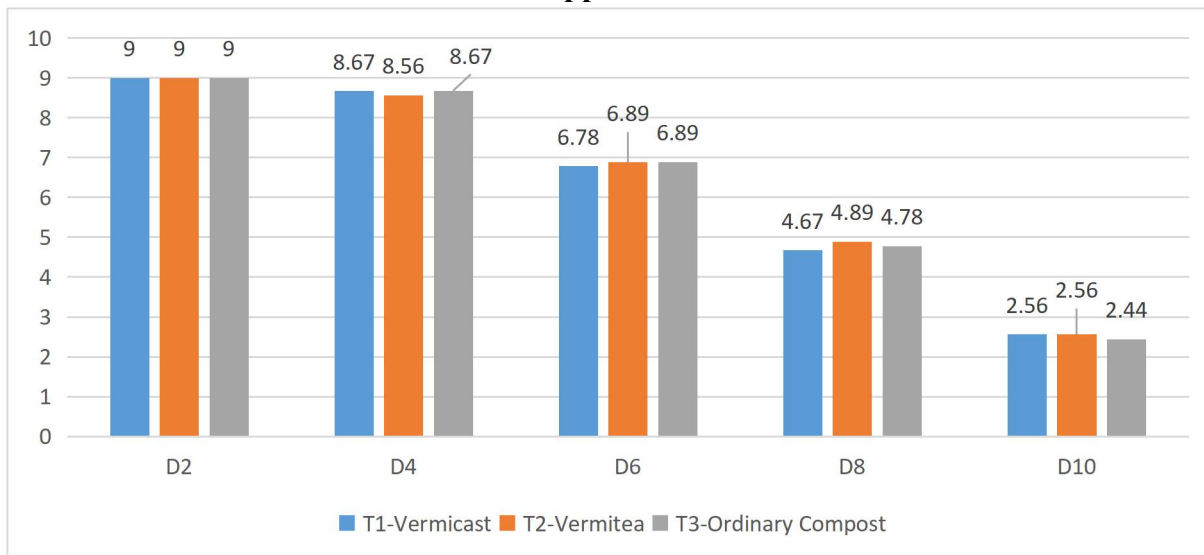


Figure 1. A graphic chart of Visual Quality Rating (VQR) of sweet corn applied with different soil ameliorants.

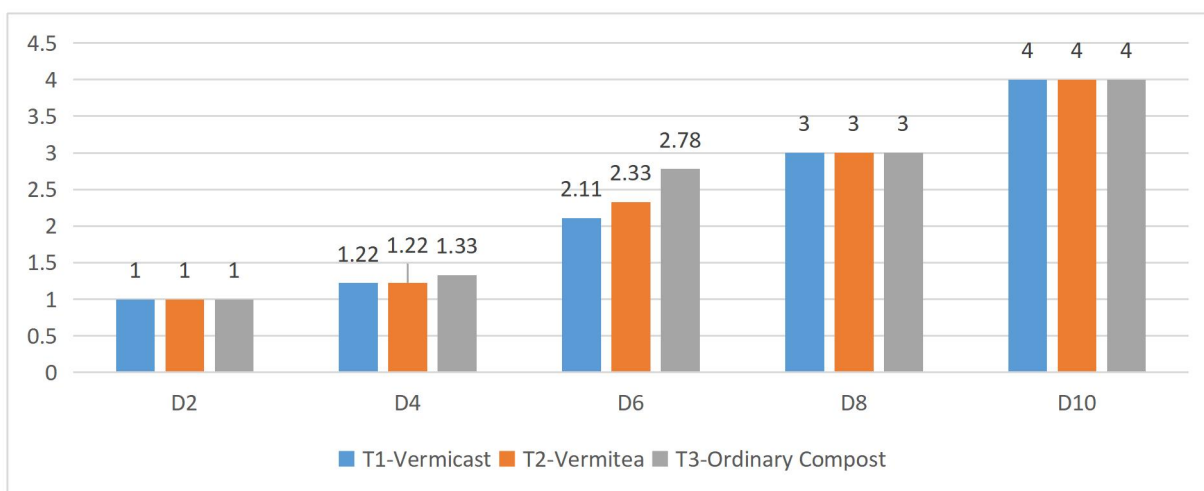


Figure 2. A graphic chart of Shriveling rate of sweet corn applied with different soil ameliorants.

Treatments	Initial pH ^{ns}	Final pH ^{ns}	Initial TSS(Brix) ^{ns}	Final TSS(Brix) ^{ns}
T1 - Vermicast	6.88	6.03	6.63	5.80
T2 - Vermitea	7.02	6.01	5.83	5.57
T3 - Ordinary compost	6.93	5.76	6.17	5.67

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Table 1. Initial pH, Final pH, Initial TSS and Final TSS of sweet corn applied with different soil ameliorants.